

- ① 오답.md 어제 오답부터 재풀이 ② 신규는 공부계획 ID대로 ③ 채점: solve-exam-prep 로 풀이 대조

떠올릴 개념

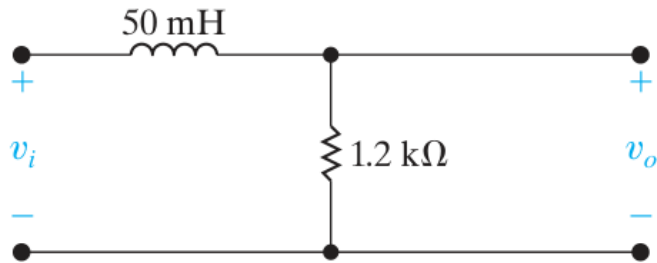
- cutoff $\omega_c = 1/RC$ (RC), R/L (RL)
- $|H(j\omega)|$ · 위상, $\omega \rightarrow 0/\infty$ 극한

문항 목차 (총 4문항)

- [1] 14.1 — 필터·공진
- [2] 14.4 — 필터·공진
- [3] 14.11 — 필터·공진
- [4] 14.13 — 필터·공진

— 풀이 —

- 14.1** a) Find the cutoff frequency in hertz for the RL filter shown in Fig. P14.1.
- b) Calculate $H(j\omega)$ at ω_c , $0.125\omega_c$, and $8\omega_c$.
- c) If $v_i = 20 \cos \omega t$ V, write the steady-state expression for v_o when $\omega = \omega_c$, $\omega = 0.125\omega_c$, and $\omega = 8\omega_c$.

Figure P14.1

— 풀이 —

14.4 A resistor denoted as R_L is connected in parallel with the capacitor in the circuit in Fig. 14.7. The loaded low-pass filter circuit is shown in Fig. P14.7.

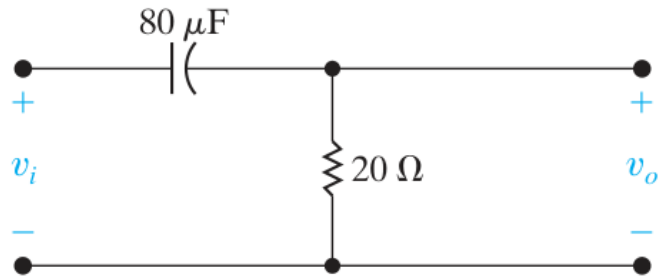
594 Introduction to Frequency Selective Circuits

- Derive the expression for the voltage transfer function V_o/V_i .
- At what frequency will the magnitude of $H(j\omega)$ be maximum?
- What is the maximum value of the magnitude of $H(j\omega)$?
- At what frequency will the magnitude of $H(j\omega)$ equal its maximum value divided by $\sqrt{2}$?
- Assume a resistance of $320\ \Omega$ is added in parallel with the $5\ \mu\text{F}$ capacitor in the circuit in Fig. P14.4. Find ω_c , $H(j0)$, $H(j\omega_c)$, $H(j0.2\omega_c)$, and $H(j5\omega_c)$.

Figure P14.4

— 풀이 —

- 14.11** a) Find the cutoff frequency (in hertz) for the high-pass filter shown in Fig. P14.11.
- b) Find $H(j\omega)$ at ω_c , $0.125\omega_c$, and $8\omega_c$.
- c) If $v_i = 75 \cos \omega t$ V, write the steady-state expression for v_o when $\omega = \omega_c$, $\omega = 0.125\omega_c$, and $\omega = 8\omega_c$.

Figure P14.11

14.13 Using a 100 nF capacitor, design a high-pass passive filter with a cutoff frequency of 300 Hz.

**DESIGN
PROBLEM**

- a) Specify the value of R in kilohms.
- b) A $47\text{ k}\Omega$ resistor is connected across the output terminals of the filter. What is the cutoff frequency, in hertz, of the loaded filter?